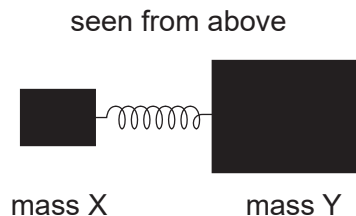


PAPER 1 / QUESTION 119

3 May 2021 / TZ1 / HL Paper 1 / original question 6 / 2221-6507

- 119.** Masses X and Y rest on a smooth horizontal surface and are connected by a massless spring. The mass of X is 3.0 kg and the mass of Y is 6.0 kg. The masses are pushed toward each other until the elastic potential energy stored in the spring is 1.0 J.



The masses are released. What is the maximum speed reached by mass Y?

- A. 0.11 ms^{-1}
- B. 0.33 ms^{-1}
- C. 0.45 ms^{-1}
- D. 0.66 ms^{-1}